

# Automated News Aggregator

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The way readers consume news has evolved as a result of the rise of the internet and social media. Over the last two decades, newsrooms have expanded their operations online, and their stories are now published on social media, online web portals, and/or mobile applications. The internet has democratized and facilitated journalism, while social media has made it easier to exchange and spread news [1]. Although this is generally positive, there may be certain instances where it has a detrimental impact. If the news is biased or inaccurate, it may distort the public's perception of critical issues [2]. Since articles have very similar subject matter, and most of the time originate from the same press release, this seemed a good opportunity to transfer the problem of Plagiarism Detection. We referenced [3] to investigate suitable clustering techniques. Works of Bias Detection and Mitigation has been investigated, particularly in the lens of Gender Bias, Political Bias and Racial Bias, using CNN Classifier, Multilayer Perceptron (MLP) Model, and Transformer Classifier respectively [4], [5] and [6]. Different summarization works have been considered, including the best way to organise articles for summarization [7], and the use of optimisation techniques in order to enhance the summarization process [8]. Therefore our proposed solution is an online platform in which articles related to the same subject published by multiple newsrooms are aggregated into one article with minimal bias. This project aims to work in a transparent manner and make use of Explainable AI to show the user how decisions are being made. Currently, similar systems just group similar articles and stories together [9] and [10]. This aims to take it a step further by actually aggregating the article's content, all the while producing as little bias as possible.

Additional Key Words and Phrases: aggregator, news, bias, clustering, summariser

## 1 INTRODUCTION

With the rise of the internet and social media, the way readers consume news has changed. In the past 20 years, newsrooms have shifted their operations online and now publish their articles on social media, online web portals and/or mobile applications. The internet has democratised and facilitated journalism, while social media made it easier for news to be shared and disseminated [1]. While this is overall very positive, there may be some cases where its effects leave a negative impact. If the news is biased or incorrect, it may impair the public's perception of crucial problems [2].

Furthermore, if readers only follow newsrooms with which they normally agree, they risk becoming trapped in echo chambers [11]. An echo chamber is a term used to describe a phenomenon in media in which a person is exposed to just their viewpoints, with no challenge or diversity of perspectives [12] and [11]. This was already a problem in traditional media, but it has now been exacerbated by the number of newsrooms and content available.

Therefore, we propose this aggregator, which aims to tackle the reality of multiple news outlets and the various perspectives and/or biases they infer in their articles. The objective is to present this impartially and transparently - such that the reader can identify the sources and decisions taken by the system. The system should be able to detect and mitigate bias and aggregate the sources' articles

into one article containing all the relevant and important information. Maltese Local news will be taken as a case study for this research. First, the dataset is created, by scraping the major Maltese local news sources. The articles are then processed, embedded and then clustered. Text Analysis, such as Sentiment Analysis and Bias Identification, is then performed on the clustered articles. The articles will then be aggregated into one article, will then undergo analysis to see whether the output is correct and alleviated from bias.

## 2 AIMS AND OBJECTIVES

The objectives of this project are two-fold. (1) Can we aggregate news articles from different sources into one article? and (2) Is it possible to produce an article based on biased material with minimal bias? These are to be implemented while keeping the process transparent to the user as much as possible. The aggregated article will allow the user to view an article with a better understanding due to a fuller perspective on the topic. The use of Explainable AI, allows the user to trust this system's results and understand the judgement taken by the algorithm. Also, this will serve as a means of verification and feedback, in the case where the aggregation software makes a wrong judgement, as it would be easily identifiable by the user.

## 3 BACKGROUND RESEARCH AND LITERATURE REVIEW

### 3.1 Natural Language Processing (NLP)

NLP is a branch of AI that deals with text analysis, understanding, and generation. NLP tasks initially solved translation problems then and progressed to information extraction and summarization [13]. Machine Learning breakthroughs have accelerated the growth of NLP.

The technique of gathering information and comprehending unstructured text is known as *text analysis*. [14] explain different concepts within text analysis such as Information Extraction, Topic Extraction, Summarisation, Clustering and Deep Learning among others, and the relevant topics of this research will be discussed further.

*Information Extraction* finds and extracts keywords and phrases from the text. *Named Entity Recognition (NER)*, in which features such as names of locations, people, and organizations are identified and extracted from text [14] and [15], and *Coreference Resolution*, in which synonyms of words and phrases are replaced by their first occurrence, are two common techniques used in information extraction [16].

A common NLP technique that is used to associate text with a particular predefined set of topics and subjects is known as *Topic Classification*. One approach of classification is to check the words from within a text, alongside their synonyms and related words, against the list of predefined terms. If they exist, the occurrence is counted, and a category is assigned based on the frequency of the

terms [14]. Deep Learning approaches also exist and perform very well [17].

With the vast volume of information available on the internet, text summarisation is becoming increasingly popular. Automatically summarising the likes of articles, reviews and academic papers is becoming a highly sought-after service since doing it manually is not always feasible. In essence, summarization is the process of condensing the full-length text into a smaller piece of text containing most of the important points and meaning [14] and [18]. There are two main streams for summarization; Extractive and Abstractive Summarisation. *Extractive Summarisation* is the process of identifying and extracting important sentences within the text and compiling them. Whereas *Abstractive Summarisation* creates a summary from the original text using unique words and sentences. Its goal is to produce a summarised piece of text in the same way as a person would.

Having multiple documents, and grouping them together according to particular attributes, is known as *clustering*. There are various techniques to cluster documents together [3]. This is discussed further in section 3.3. Similarly, identifying certain topics or concepts, related to a particular document, or trying to find different relationships, is called *Concept Linkage*. It aims to make associations between different documents and outline new and possibly unforeseen connections [14] and [19].

### 3.2 Clustering

Plagiarism Detection has been considered for this project as an approach to cluster similar news articles.

Plagiarism Detection is the process of identifying near-duplicate documents [3]. Therefore this process is viewed as a similar problem to identifying similar articles. Since most newsrooms discuss very similar and current topics and sometimes even originate from the same press release, therefore similarities between their articles are bound to be present.

The Plagiarism Detection problem aims to find the documents from which a document  $D_{Query}$  copied. Plagiarism Detection can vary in complexity, especially in instances where Paraphrase Plagiarism takes place. Paraphrasing is the instance when the author changes the original text's structure and makes use of other techniques to alter the text such as Semantic Equivalence, Discursive Changes and Morphological Substitutions [3]. This text looks different but maintains roughly the same meaning. Furthermore, text can be intentionally made unclear or obscure, for the sake of being different from the original.

[3] discuss that with techniques that make use of Bag of Words (BOW) or Vector Space Models (VSM), words are considered at a very shallow level since the meaning of the word is not taken into account. To face this shortcoming, the authors approach the problem by creating an embedding of the word to word vectors, using Word2vec. This embedding outlines the semantic relationship between the words and thus, these words can be clustered. These newly formed clusters are said to be concepts - a group of words that have similar meanings.

Before clustering the vocabulary into concepts, Principal Component Analysis is performed on the word vectors, to decrease the

dimensionality, since the authors noted that in some cases the vectors were too large for clustering. The cosine function was used as a distance metric between the word vectors. Thus, the Spherical K-Means clustering algorithm was used, since it makes use of the cosine function.

Once concepts are drawn for each document,  $D_{Query}$  is filtered against 'reference documents' (documents with similar concepts). The next step is identifying, the exact sentences which are plagiarised. This is done by first comparing the similarity of the query document's paragraphs to the reference documents' paragraphs. A similarity threshold is chosen, and if the similarity of a pair of paragraphs exceeds it, its sentences are considered for the next step of similarity. The sentences within the chosen paragraphs are compared once more. The most similar sentences are then said to be plagiarised.

While the application of semantic understanding improves on past work with BOW and VSM, it still has some shortcomings [3]. Mainly, this approach doesn't take the semantics of the document, paragraph and sentences into account. If a sentence is negated by the use of the word 'not', the similarity still results to be high, therefore cases like this result in a false positive prediction. We aim to tackle this problem by using an Entity Semantic Analyser. This will allow the algorithm to understand whether the query document is contradicting the reference documents or plagiarising. Furthermore, rather than using Google's Word2vec, we will be using Google's Universal Sentence Encoder, which is built to handle sentences, phrases and paragraphs. This provides a better result in the encoding stage and therefore can extract semantic similarity more accurately and efficiently.

### 3.3 Bias Detection and Mitigation

Detecting Bias is a core part of this project. Here, three papers investigating three kinds of biases are discussed, showcasing different approaches to achieve a similar goal.

A study by [4], investigated *gender bias*, with respect to academic texts. The study found a relationship between the author's age and gender and the gender inclusivity in their writing. This was done by training a CNN on doctoral theses written in Spain between 1974 and 2020, which composed a dataset of around 100,000 theses. Data labelling was carried out by experts on sub-sentences which were reviewed by two people, classifying them into two types, 'Inclusive' or 'Non-Inclusive'. A third person categorized any conflicting types.

Analyses were drawn on the reliability and validity of the algorithm's output and to draw patterns and trends on what were the factors that related to the inflicted bias. The algorithm scored an accuracy of 94% and a precision of 92%. The research concluded that there is less gender awareness in older males than in younger candidates in general. Overall, women are more aware of gender awareness. This study shows very promising results for a CNN classifier with a well-labelled dataset.

[5], aimed to tackle *political bias* detection. The approach was taken using a MLP Model. The classifier was built to classify Ideological Components within an article, classifying whether the sentences were Conservative, Liberal or Neutral. The classifier was trained

on the Ideological Books Corpus, which is made up of over 4,000 annotated sentences on Political Ideologies.

This implementation achieved an F1 score of 81% on the test set but didn't achieve satisfactory results for real-life applications, this was attributed to the use and likes of metaphors and negative phrases. This shows that the learning done was shallow and the text wasn't understood by the classifier, another factor could also be the embedding of the words. The embedding was done using fastText, in which words were seen as n-grams made up of characters. This did not cater for semantics in the embedding, and thus couldn't handle semantic structures like negation and metaphors [5] and [3].

[6] aims to detect hate speech and mitigate *racial bias* on Social Media. This is attempted using the pre-trained BERT model. The authors note that while the other supervised systems perform well, the use of good, impartial and sufficient datasets still remains an issue.

The study is done primarily on the datasets of the works of [20] and [21]. The former tackled hateful, and offensive texts whereas the latter focused on Racist or Sexist texts. The baseline for the evaluation is the F-Score of their work and models.

The study bias detection and mitigation models used the pre-trained BERT Model. The input sequences were split into subwords using WordPiece tokenisation. The study is done with the BERT model on the different fine-tuning strategies. The strategies include BERT, BERT + Nonlinear Layers, BERT + bi-LSTM and BERT + CNN. All the fine-tuning strategies but one (BERT+nonlinear classifier) outperformed the pre-existing approaches on the datasets. The best fine-tuning strategy was the BERT Model with a CNN Layer.

The macro averaged F1-score will be used to evaluate the performance of the models.

The study noted that sentences with insulting and derogatory words were easy to catch, whereas sentences with inferred/implicit racism were almost always misclassified. Although in some cases, even humans may have a hard time classifying it correctly. The authors also discussed the possibility of the annotators being over sensitive with certain text since in certain cases where the sentence contained an offensive word but was not particularly hateful or offensive, it still got classified as hate.

This study proved that the pre-trained BERT Model (along with the fine-tuning strategy), minimised the data needed to train.

After seeing a significant improvement in bias detection, [6] started another investigation. Is it possible to use the same model and apply re-weighting schemes and mitigate the bias? This was approached by first, examining the bias in the dataset, and identifying what was causing it, then reweighting the samples in the training and validation set, after optimising the bias in the training set, the re-weighted scores for each sample is retrieved and fed into the BERT model with the new training and validation sets. The loss function of the classifier is updated with the new scores during the fine-tuning.

The authors concluded through experiments that the model managed to alleviate racial bias. They still noted that the system is by no means perfect and still had some traces of bias, but it is considered as an advancement in this field.

With these three studies, we outlined three different techniques of detecting bias. Using a CNN classifier [4], using an MLP Classifier

[5], and finally using BERT [6]. Other works also include phrase statistics[22].

### 3.4 Aggregation and Summarisation

Summarisation is the process of reducing the length while preserving the main points and overall meaning of a document [14]. According to a study by [7], the organization of articles is a major component in effective summarization. The research evaluates how different hierarchical structures are used and how effective they are at summarizing documents. The number of documents, time interval, and event subjects were all compared in three clustering structures. According to the findings, hierarchical structure by event subjects performed best when summarizing with a high compression rate.

[8] propose two versions of text summarisers using the Genetic Algorithm with Hierarchical Clustering (GA-HC) and Particle Swarm Optimization with Hierarchical Clustering (PSO-HC).

The researchers recommend a four-step procedure. The text is first cleaned and normalized as part of the preprocessing process. The semantics of words are then extracted using a semantic model. A clustering algorithm is used to group similar sentences, and then an optimised ranking algorithm is used to rank the sentences in the document. To construct the summary, the rated sentences are normalized and merged.

Semantic extraction is carried out with the use of a distributional semantic model and statistical analysis of the words used in the context. On the distributional semantic algorithm, the researchers employed the Word2vec model. Word2vec creates vectors of words with related semantics to represent sentences. The vector of sentences is the summation of the word vectors within the sentences.

[8] make use of Hierarchical Clustering, which further supports [7]. Clustering is used to associate similar sentences within the document. All sentences are initially considered to be individual clusters, and then depending on the distance metric, are clustered together.

[8] compare 3 different techniques to rank the sentences; the traditional ranking algorithm, Genetic Algorithm (GA) and Particle Swarm Optimization (PSO).

To rank the sentences, the Ranking Algorithm is utilized, and the top n sentences are used for the summary. According to [23], there is a link between the length of a sentence and its importance in the document. Using statistical information from the document, such as term frequencies, the *TF-IDF score* selects key sentences. The *sentence position* is taken into account because the first few phrases of news items usually include the most important information. We presume that the more *noun and verb phrases* in a sentence, the more information it contains. Sentences containing *proper nouns* are also regarded to provide important information. The following GA and PSO-based methods are used to optimize the total score that results from these features. Finally, the *Cosine Similarity* is used to compare the vectors.

The GA is an algorithm modelled around the Evolution of Life in which the phenomena of natural selection is at play. The fittest individuals are selected, and their characteristics move on to the produced offspring in the next generation. Mutation and crossover

are factors that add noise to the selected characteristics and improve on find improvements.

The PSO is an algorithm based on the behaviour of the flocks of birds. Each particle has its own best position and is trying to find the best global position in the search space. The swarm collectively finds the best global position [24].

[8] used the Non-Dominated Sorting GA II to rank the sentences which is an improvement on the original GA algorithm, which resulted in faster convergence.

In summary, once the vectors are created and clustered, the vectors are passed on to be optimised by the GA-HC or PSO-HC ranking models and the top n sentences are selected and a summary is formed.

The paper tests the results by passing the summaries to ROUGE 1, 2, 3 and L. K-Means and Fuzzy C-Means were used to test other clustering techniques, and Hierarchical clustering had the best outcome. Overall the GA had the best Precision and F1 scores, whereas PSO had the best recall score for ROUGE 1 and L. However [25] states that there is a poor correlation between human judgement and ROUGE-L, especially when compared with ROUGE 1, 2 and 3 [18].

[18] created and trained a Transformer Language Model (TLM) that generated high-quality abstractive summaries of particularly long texts (including research papers and articles). The system made use of an extractive step on the text, followed by an abstractive step which proved to be very useful as this model outperformed the previous extractive summarization baselines on arXiv and Pubmed datasets. The model also outperformed the previous abstractive and mixed summarization results on the arXiv, PubMed and bigPatent datasets, with the exception of ROUGE-L. The authors comment that this is due to the lack of a copy-mechanism, it further references the works of [26] which state that ROUGE-L is particular distant from human judgement.

The two components of this paper include a hierarchical document representation model (using a bidirectional LSTM encoder-decoder), that is used to generate the extractive summary and a TLM (GPT-like transformer) that conditions the extracted sentences and the full text. The model was trained on arXiv, PubMed, bigPatent and Newsroom.

[18] concluded through the results that the extractive sentences were very helpful for the transformer. The paper noted, that the model was less likely to copy entire sentences from the inputted text. Training the model included the papers' introductions, the extracted sentences (from the extractive model), the abstract and the rest of the paper.

Finally, they remarked that evaluation of summarization tasks should cater for 'factual correctness' and 'coherency'. Moreover, they added that evaluation of summaries should be done by highly trained human evaluators, after reading the entire source material.

## 4 PROPOSED SOLUTION

This project aims to tackle the reality of multiple news outlets and their different opinions and/or spins that they infer in their articles. It should be able to detect and mitigate bias and aggregate the sources' articles into one article containing all the relevant and

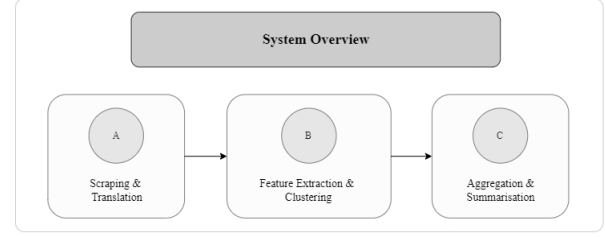


Fig. 1. Block Chart illustrating the System Overview.

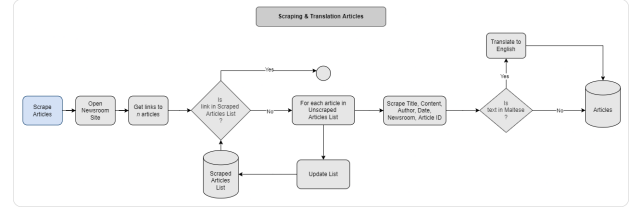


Fig. 2. Flowchart illustrating the Scraping and Translation Process.

important information. An overview of the proposed system can be seen in figure 1.

Multiple techniques and areas within NLP will be employed, including Semantic and Sentiment Analysis, Text Summarisation and Text Aggregation, while using State of the Art (SOTA) techniques. In addition, the project will be employing Explainable AI during its decision-making process, to keep the process transparent, so that the user can understand the judgement taken by the algorithm to achieve the result produced. This will be done, by showing as many small decisions and results taken along the way, such as which sentences came from which sources, which sentences differed from the rest and the sentiment and correctness scores of the generated summary. This will also enable the user to leave feedback if they identify a mistake or don't agree with certain reasoning or judgement taken during the process.

The News Aggregator first starts by Scraping the News Articles from the various sources. The project covers the 9 principal Maltese online sources: The Independent, Illum, LovinMalta, MaltaDaily, MaltaToday, Net News, NewsBook, One News, Times of Malta and TVM News. Once the articles are gathered, they will be processed and embedded. Similar news articles will be clustered together and Text Analysis (TA) will be performed on each article. TA can take various forms, ranging from Sentiment Analysis to Bias Identification [14]. Text Aggregation and Summarisation will ensue, once the analysis is complete.

### 4.1 Gather the Datasets

Scraping will be done, using Selenium. The scraper will be built to be as general as possible, such that if a news source needs to be added, it can be appended to the list of sources. The scraped contents of each article will include the: link, date and time of publication, native article ID, title, content, author, topic tags and the source. Only articles with more than 30 words will be considered in the

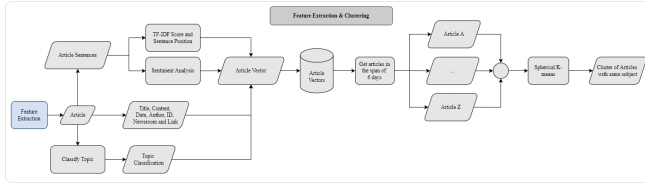


Fig. 3. Flowchart illustrating the Feature Extraction and Clustering Processes.

dataset. The text will be processed, direct quotes will be removed, as to only take into consideration the journalist's words [22].

The articles will be stored as 2 sets of datasets. The first set is organised by source, and the second by date. This is done to facilitate the performance of getting articles within a specific date range, without having to read all of the articles.

For articles written in Maltese, in this case, by Net News and One News, the content will be translated to English, using the Google Translate API, to use NLP techniques prepared/pre-trained on the English Language.

A flowchart in figure 2 can be seen illustrating the two processes

## 4.2 Identify Article Features

In this step, an Article vector is formed. The vector will be made up of the scraped content acquired in the previous step, as well as its topic, which will be classified using a Topic Classifier. Additionally, the sentences within the body of the article will be passed through an entity sentiment analyser, through which entities will be identified together with their sentiment. TF-IDF can also be used to get more statistics from the phrases used. Entity Sentiment Detection and Topic Classification can be performed using Google's Cloud Natural Language API since it provides SOTA results.

## 4.3 Cluster Similar Articles

Once the article vector is created, the article title and body will be embedded using Google's Universal Sentence Encoder (USE). The USE embeds each sentence to a 512-dimensional vector. With this embedding, Text Analysis such as Clustering and Entity Sentiment Analysis can be implemented.

To obtain different accounts of the same article written by various sources, we will assume that all of the articles regarding the subject will be published within a week of each other. Each cluster of articles relating to the same topic will be made up of at least three or more articles. We aim for the size of the cluster to be at least equal to the number of sources. If a source within the cluster returns more than one article, the most recent one will be taken into consideration since most articles are updated with the latest information. Spherical K-Means will be implemented since the cosine distance metric will be used. After this, we end up with a cluster of around  $n$  (where  $n$  is the number of news sources) articles. Figure 3 is a pictorial representation of the Feature Extraction and Clustering processes explained above.

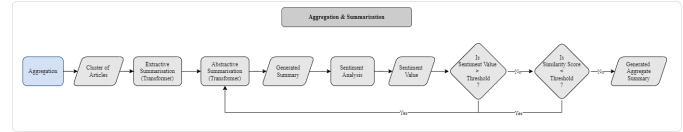


Fig. 4. Aggregation and Summarisation

## 4.4 Aggregate Articles

As figure 4 illustrates, the clustered articles will be individually summarised, using Extractive Summarisation [18]. The summary will be generated using a transformer model such as BERT, as from the works of [18] the model yielded good results. Different models may be tested to investigate which produces the most optimal result. Sentiment Analysis is carried out, this time on the summary, and an aggregate sentiment value is outputted.

To aggregate the articles, a transformer, the likes of GPT-3 will be used, which will take the extractive summaries of all the articles and output one article. GPT-3 has recently become open-source and thus, making use of the pre-trained 175-billion parameter model should promise good results. Since the model does not have an aggregator model built-in, we will try to teach it using few-shot learning [27].

The article generated will go through two checks, first for bias and then for quality assurance. Therefore, two threshold values are created, one for sentiment and the other for correctness. The sentiment threshold will be a pre-determined maximum deviation value i.e. the difference between scores cannot be greater than 0.2. Whereas the correctness threshold will be the minimum similarity score accepted. These thresholds will be evaluated against results done on different tests.

The sentiment of the article is analysed and compared to the aggregated sentiment of the original articles. If the difference in the sentiment of the generated text and original articles is beyond the sentiment threshold, the aggregate article will be regenerated. The second check is done against the extractive summaries generated in the previous step. The average similarity score between the summaries and the newly generated article is computed. If the average similarity is below the similarity threshold, then the article will be regenerated. The transformer keeps on generating text until both scores are minimally satisfied (according to the thresholds). A title will also be generated using the transformer.

## 4.5 Perform Analysis on Aggregated Article

The aggregated article undergoes the same analysis as the original articles, this time adding correctness as a score. Correctness is measured by taking the average of the similarity between the generated text and each article, and the average is calculated.

## 4.6 Publish MVP

This process will be hosted on a website, taking as input a query or a link to an article (from the 9 sources). Additionally, the homepage will have a dashboard with the trending articles and topics. The website will illustrate clearly the differences between the sources and the generated aggregation and will allow the users to rate the

article based on correctness, aggregation and bias mitigation. This will be considered as ongoing feedback, to improve the system.

## 5 EVALUATION PLAN

Net News and One News are both funded by Political Parties, TVM News is funded by the government, and the other media outlets are independent newsrooms. Net News is funded by the Nationalist Party whereas One News is funded by the Labour Party, it is assumed by the general public that their articles have some sort of bias favouring their party while having a bias against each other. It will be interesting to note the political bias scores of these outlets in an objective way since a lot of speculation is always assumed. Political Bias is the act of changing or showing certain information to make a political position or entity seem attractive or bad, depending on the agenda of the author.

To test the different forms of biases of the articles, we will perform different tests. Some tests will be automated, whereas others will be hand-picked, in order to investigate specific trends.

Within a given period, for example, 7 days, we will get the most popular topic i.e. the biggest cluster. Entity sentiment analysis will be carried out on the top  $n$  entities as well as further statistical analysis will ensue, such as word frequency and collocation. These results will be compared with all of the newsrooms.

We will compare the frequency of certain hand-picked charged topics, against the different newsrooms. Too much or a lack of an adequate amount of coverage is a sign of bias since the reader is either getting inundated or not educated enough on the subject at hand. Therefore, the frequency of the topics will be compared for all the newsrooms, to see whether there are certain trends relating to topics.

Another test that will be done, is a sentiment analysis score on hand-picked uncharged topics. If the uncharged topic such as ‘MoneyVal’ (which is a supposedly regular and neutral process) is put under certain particularly positive or negative lenses, then there is some sort of bias at play, such as potentially politicising unpolitical concepts.

With regards to political bias, there is a lot of literature related to American Politics [5], [22], in which there exists Left, Right and Centre leaning parties. Maltese Politics is mostly central, and there are only minor differences between ideologies, it is not particularly significant to be cast as left-wing or right-wing. Thus when relating to political bias it won’t be considered as a three-labelled problem (or 5-labelled, since some literature also includes right-centre, left-centre) but rather a binary classification.

The online interface will provide space for each article to receive feedback from the user. The feedback will be based on three main factors, correctness, aggregation and bias mitigation. Each factor will be scored anonymously out of 5 stars. The results will be monitored to see improve the system accordingly.

## 6 CONCLUSION

The consequences of having a numerous amount of newsrooms and in turn news articles are mixed. On the one hand, it delivers more complete and diversified perspectives, as well as a purportedly more accurate picture of the truth. But on the other hand, it may

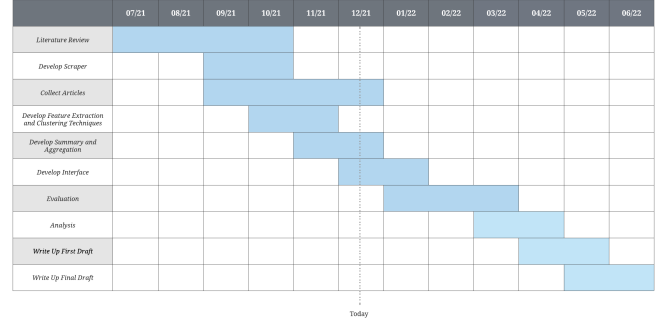


Fig. 5. Gantt Chart for the proposed methodology.

become overwhelming [28] or even confusing for the reader, if the articles are presenting the same story but with different views due to possibly inferred unknown and hidden biases.

This study will develop a system that aggregates articles’ information which may aid the reader in gaining a better and more complete picture of the present local events that the articles are attempting to portray. Moreover, this is done while highlighting the detected biases and alleviating bias from the generated aggregated article. This study makes substantial use of NLP, which encompasses a wide range of tasks, including Text Analysis. Text Analysis allows us to extract information, perform semantic analysis, and construct summaries. The paper investigated bias detection and mitigation, clustering algorithms based on plagiarism detection, and summarization.

Figure 5 shows the estimated timeline for the proposed methodology. As seen in the figure, collection of articles will be done throughout, since articles are posted daily. Evaluation is planned on three months, since some scores will be provided by the users. This will ensure enough time for collection. Once the evaluation phase is complete, time is allocated for analysing the system and its evaluation.

This paper outlined the proposed solution for the automated news aggregator. We describe how the dataset is gathered, how article features are identified, and how articles will be clustered. Then we move on to summarization and aggregation, and lastly, the analyses of the aggregated article. The end product will be an online interface that allows users to preview and search for article topics. Upon selecting the article or topic, the aggregated article will be displayed, alongside notes on the sources and the overall bias. This representation of local news should help the users understand better the full story while consuming minimal bias.

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